Revealing Key Details to See Differences: A Novel Prototypical Perspective for Skeleton-based Action Recognition

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Abstract

Skeleton-based action recognition has become a strong alternative to RGB video modeling, but it still struggles on actions whose differences are confined to a few joints or subtle local motions. This failure mode is especially visible on fine-grained classes, where adaptive graph convolutional networks often learn useful coarse topologies yet under-emphasize the small discriminative details that separate confusing categories. We present ProtoGCN, a prototype-driven graph convolutional framework that reconstructs learned motion topologies from a compact memory and uses the reconstruction to emphasize reusable yet discriminative joint-relation patterns. ProtoGCN augments each graph convolutional block with Motion Topology Enhancement, which combines a learnable base topology with correlation- and difference-based adaptive relations, and it further regularizes the reconstructed topology representation using a class-specific contrastive objective with momentum-updated class centers. On the provided benchmarks, ProtoGCN achieves 93.8% and 97.8% on NTU RGB+D cross-subject and cross-view, 90.9% and 92.2% on NTU RGB+D 120 cross-subject and cross-setup, 51.9% Top-1 and 75.6% Top-5 on Kinetics-Skeleton, and 95.9% on FineGYM. Ablations show a consistent contribution from each component, with the largest gains appearing on the most difficult NTU-120 classes.

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1. Introduction

Skeleton-based action recognition represents an action sequence as human joint trajectories, offering robustness to appearance variation and a compact input structure compared with RGB-based alternatives [7, 18, 24]. The graph formulation is particularly natural: joints are nodes, bones define edges, and graph convolutional networks (GCNs) aggregate motion information over space and time. This line of work has produced strong progress, from fixed-topology models [24] to adaptive and multi-scale architectures [2, 11, 12, 15].

Despite this progress, fine-grained discrimination remains difficult. Many action pairs differ only by subtle motions around a small subset of joints. Figure 1 illustrates a representative example: *Writing* and *Typing on a Keyboard* involve similar global posture, while the actual distinction is concentrated in limited hand-arm relations. When appearance cues are absent, the classifier must rely almost entirely on nuanced topology changes. Existing GCNs can adapt their adjacency matrices, but the learned relations are still often dominated by coarse or shared patterns, which makes confusing classes harder to separate.

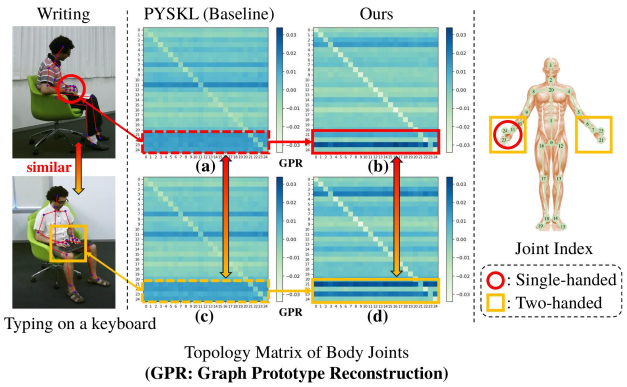

Figure 1. Similar actions can share global pose while differing in a few discriminative joint relations. The provided example compares *Writing* and *Typing on a Keyboard*.

We address this limitation from a prototypical viewpoint. Our key observation is that many useful motion relations recur across samples, but not every relation should contribute equally to every action. Instead of using only a directly learned topology, we reconstruct the learned topology features from a small prototype memory. This encourages the model to represent motion as a composition of reusable relation patterns while still producing sample-specific emphasis through soft addressing. The resulting representation makes fine-grained differences more explicit and provides a natural target for discriminative regularization.

Based on this idea, we propose **ProtoGCN**, a prototype-enhanced GCN for skeleton action recognition. ProtoGCN contains three coupled components. First, a **Motion Topology Enhancement** (MTE) module enriches each GCN layer with two adaptive relation terms on top of a learn-

able universal topology. Second, a **Prototype Reconstruction Network** (PRN) reconstructs the final learned topology tensor from a memory of prototypes through soft assignment. Third, a **Class-Specific Contrastive Learning** (CSC) loss regularizes the reconstructed topology representation using momentum-updated class centers, pulling samples toward their class prototype while separating them from other classes. Figure 2 gives an overview.

Our contributions are threefold. First, we introduce a prototype-based perspective for skeleton GCNs that reconstructs learned motion topologies instead of using only direct feature propagation. Second, we design a practical architecture that combines topology enhancement, prototype reconstruction, and class-specific contrastive supervision in a single end-to-end framework. Third, on the supplied experiments, ProtoGCN improves over strong recent baselines on NTU RGB+D [16], NTU RGB+D 120 [14], Kinetics-Skeleton [9], and FineGYM [17], with especially strong gains on difficult classes.

2. Related Work

2.1. Skeleton GCNs and topology learning

Modern skeleton recognition methods largely build on GCNs that model joint dependencies directly on spatio-temporal graphs [24]. Subsequent work improved the backbone through adaptive topologies [4, 12, 18], enlarged receptive fields with multi-scale operators [3, 15], or refined topology generation using channel-wise and hierarchical decompositions [2, 11, 27]. Recent models also incorporate stronger priors or spatio-temporal graph construction strategies [5, 10, 21, 25]. These methods substantially improve recognition accuracy, yet their topology learning is still primarily driven by direct feature adaptation. ProtoGCN differs by explicitly reconstructing topology features from a learnable prototype memory, targeting fine-grained relation patterns that recur across samples.

2.2. Discriminative supervision for confusing actions

Several works try to improve class separability beyond standard cross-entropy. FR-Head adds auxiliary refinement and contrastive supervision to emphasize ambiguous samples [26]. SkeletonGCL introduces graph-level contrastive learning across sequences with memory banks [8]. GAP injects language-derived action semantics into skeleton recognition [20]. These approaches show that additional discriminative objectives help, especially when classes are visually or structurally similar. Our method is related in spirit but operates on reconstructed topology descriptors rather than directly on pooled action features, and the memory is maintained at the class-center level instead of storing instance queues.

2.3. Alternative skeleton representations and multi-stream modeling

Beyond pure GCN design, strong performance also comes from robust training recipes and complementary skeleton streams. PYSKL unifies several practical design choices and provides competitive skeleton baselines [6]. PoseConv3D revisits the representation itself by replacing graph sequences with 3D heatmap volumes [7]. Fine-grained action recognition on datasets such as FineGYM is often evaluated alongside RGB methods such as I3D, TSN, TSM, and LT-S3D [1, 13, 19, 22], as well as skeleton-specific pre-training methods such as SkeletonMAE [23]. ProtoGCN is complementary to the common multi-stream setting: the provided configs support six input streams, and the best reported results are obtained through six-modality ensembling.

3. Method

3.1. Preliminaries

Let a skeleton sequence be represented as $\mathbf{H}^{(0)} \in \mathbb{R}^{N\times C\times T\times V}$, where N is the batch size, C is the channel dimension, T is the number of frames, and V is the number of joints. ProtoGCN follows a standard stacked GCN-TCN design, using ten graph-convolutional stages with multi-scale temporal convolutions. The backbone outputs both an action feature tensor for classification and a reconstructed topology descriptor for contrastive regularization.

Each graph block starts from a learnable base topology $\mathbf{A}_0$ and computes an adaptive topology for feature aggregation. Instead of stopping there, ProtoGCN explicitly reconstructs the learned topology using a prototype memory and optimizes the reconstruction with class-specific contrastive supervision.

3.2. Motion Topology Enhancement

The MTE module is inserted into each graph convolution block. Given an input feature map, ProtoGCN first projects it into K heads and then builds two adaptive relation terms after temporal averaging. Following the implementation, one branch uses projected feature differences and one branch uses projected feature correlations. We denote them as $\mathbf{A}_{\text{inter}}$ and $\mathbf{A}_{\text{intra}}$:

$$\mathbf{A}_{\text{inter}} = \tanh(\mathbf{Q}_1 - \mathbf{Q}_2), \tag{1}$$

$$\mathbf{A}_{\text{intra}} = \text{Softmax}(\mathbf{Q}_1^\top \mathbf{Q}_2). \tag{2}$$

The enhanced topology is then

$$\widetilde{\mathbf{A}} = \mathbf{A}_0 + \alpha\mathbf{A}_{\text{inter}} + \beta\mathbf{A}_{\text{intra}}, \tag{3}$$

where α and β are learnable scaling parameters. The resulting topology is channel-aware and sample-adaptive. In

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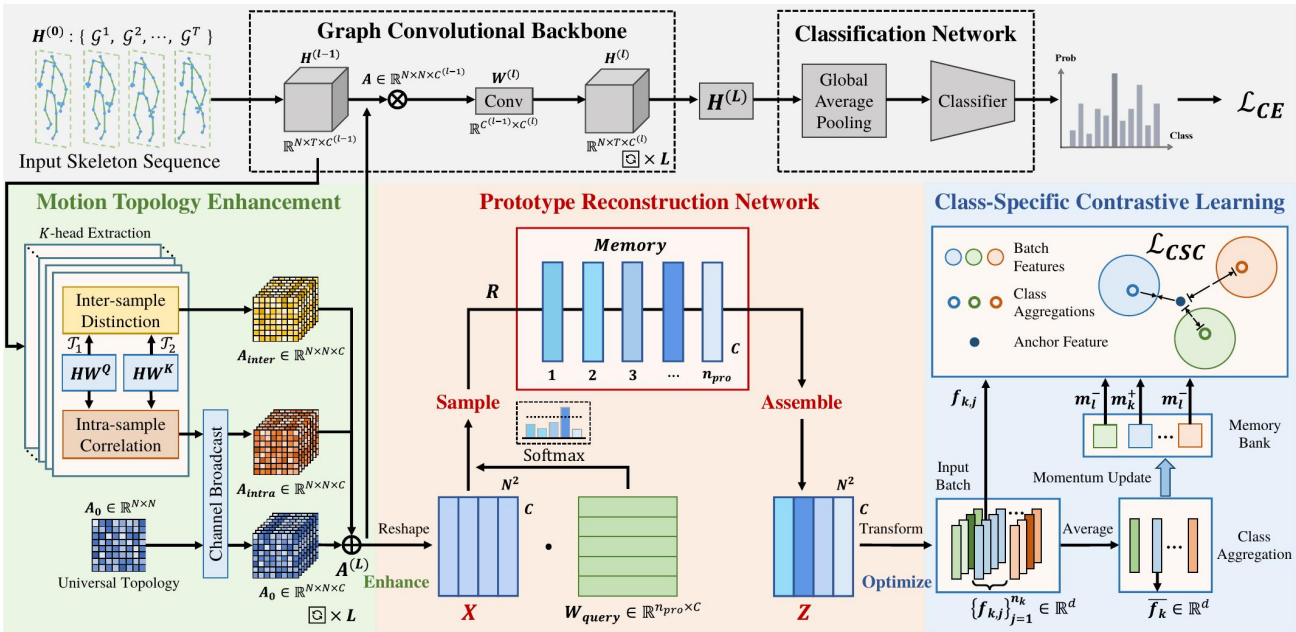

Figure 2. Overall architecture of ProtoGCN. A GCN backbone with Motion Topology Enhancement produces feature maps and learned topology tensors, the Prototype Reconstruction Network reconstructs the topology from a prototype memory, and Class-Specific Contrastive Learning regularizes the reconstructed representation using momentum-updated class centers.

practice, the correlation term contributes more than the difference term, but both are useful, as confirmed in Table 4.

This design serves two roles. First, it retains a shared topology prior through $\mathbf{A}_0$, which stabilizes learning. Second, it injects relation patterns that are more sensitive to local motion details than a fixed graph. The backbone uses $\widetilde{\mathbf{A}}$ for message passing and also exposes the final learned graph tensor to the prototype module.

3.3. Prototype Reconstruction Network

After the last GCN block, ProtoGCN extracts the learned graph tensor and averages over persons, yielding a sample-specific topology representation of shape $C \times V \times V$. We flatten the joint-pair dimension and obtain

$$\mathbf{X} \in \mathbb{R}^{V^2 \times C}. \tag{4}$$

The PRN reconstructs $\mathbf{X}$ from a prototype memory via soft addressing:

$$\mathbf{Q} = \mathrm{Softmax}(\mathbf{X}\mathbf{W}_q), \tag{5}$$

$$\mathbf{Z} = \mathbf{Q}\mathbf{W}_m, \tag{6}$$

where $\mathbf{W}_q \in \mathbb{R}^{C \times n_{\text{pro}}}$ produces soft assignment over n_{pro} prototypes and $\mathbf{W}_m \in \mathbb{R}^{n_{\text{pro}} \times C}$ maps the prototype mixture back to the topology feature space. In the implementation, this is realized by two linear layers without bias and a dropout layer.

The reconstructed feature $\mathbf{Z}$ is reshaped back to a topology tensor and further transformed by a 1×1 convolution, batch normalization, and ReLU. The final topology descriptor is obtained by channel averaging. This descriptor is not used for the primary classifier; instead, it acts as a discriminative summary of the learned motion topology and becomes the target of the contrastive loss. This separation is useful because the classifier can retain a conventional pooled action representation while the prototype pathway focuses specifically on relation structure. In effect, PRN behaves like a soft dictionary of relation templates. Samples from similar actions can reuse part of the same prototype basis, while the assignment weights still preserve class-dependent structure. This is a better match to skeleton recognition than forcing every sample to learn a fully independent topology tensor from scratch.

3.4. Class-Specific Contrastive Learning

The CSC objective operates on the reconstructed topology descriptor. Let $\mathbf{g}_i$ be the reconstructed topology vector for sample i, and let $\mathbf{y}_i$ be its label. ProtoGCN first maps $\mathbf{g}_i$ to a lower-dimensional embedding:

$$\mathbf{f}_i = \phi(\mathbf{g}_i), \tag{7}$$

where ϕ is a learned linear projection. For each class, the model stores a momentum-updated center in a memory bank. Importantly, the update uses only samples whose predicted class matches the ground truth and whose prediction

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confidence exceeds a threshold. This avoids contaminating class centers with low-confidence or incorrect examples early in training.

Let $\bar{\mathbf{f}}_c$ denote the center for class c. The center update follows a moving average:

$$\bar{\mathbf{f}}_c \leftarrow m\bar{\mathbf{f}}_c + (1-m)\,\mathrm{Avg}\{\mathbf{f}_i : y_i = c,\ \text{valid}\}, \tag{8}$$

where m is the momentum coefficient. Similarity scores between normalized features and class centers are computed with temperature τ,

$$s_{ic} = \frac{\langle \hat{\mathbf{f}}_i, \hat{\bar{\mathbf{f}}}_c \rangle}{\tau}, \tag{9}$$

and a cross-entropy loss over these scores yields the class-specific contrastive loss $\mathcal{L}_{\text{CSC}}$.

3.5. Overall objective

ProtoGCN is trained with standard classification loss and CSC regularization:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CSC}}. \tag{10}$$

The classification branch uses global average pooling on the backbone feature map followed by a linear classifier. The CSC term acts on the reconstructed topology descriptor. This combination is simple but effective: $\mathcal{L}_{\text{CE}}$ optimizes category recognition, while $\mathcal{L}_{\text{CSC}}$ explicitly increases class compactness and separation in the topology space reconstructed from prototypes.

4. Experiments

4.1. Datasets and implementation details

We evaluate ProtoGCN on four standard benchmarks: NTU RGB+D 60 [16], NTU RGB+D 120 [14], Kinetics-Skeleton [9], and FineGYM [17]. The reported protocols follow the supplied tables: cross-subject (X-Sub) and cross-view (X-View) on NTU RGB+D, cross-subject (X-Sub) and cross-setup (X-Set) on NTU RGB+D 120, Top-1 and Top-5 accuracy on Kinetics-Skeleton, and classification accuracy on FineGYM.

The released configs train all models for 150 epochs with SGD, Nesterov momentum 0.9, weight decay 5×10^{-4}, and cosine learning-rate decay. All datasets use 100-frame sampling. For NTU, the pipeline applies 3D pre-normalization and random rotation, with spatial flipping enabled in the NTU-120 configs. Kinetics-Skeleton and FineGYM use decoded 2D poses, horizontal flipping, and the provided Kinetics-style skeleton transform. The backbone uses a random-initialized learnable graph with eight filters. For single-stream models, the joint stream uses a larger prototype bank than the other streams in the released configs, and the best final results are obtained through six-modality ensembling.

4.2. Comparison with prior work

Table 1 summarizes the main results. ProtoGCN is competitive even in smaller ensembles and reaches the best reported accuracy across all listed benchmarks with six-modality fusion. On NTU RGB+D, ProtoGCN achieves 93.8% X-Sub and 97.8% X-View, improving over strong recent methods such as HD-GCN [11], JT-GraphFormer [25], and BlockGCN [27]. On NTU RGB+D 120, ProtoGCN reaches 90.9% X-Sub and 92.2% X-Set, exceeding the compared baselines including InfoGCN [5], HD-GCN [11], and BlockGCN [27].

The Kinetics-Skeleton results show that the method also transfers to noisier 2D pose inputs. ProtoGCN attains 51.9% Top-1 and 75.6% Top-5, surpassing PYSKL [6] and DS-GCN [21] in Top-1 accuracy among the listed methods. On FineGYM, Table 2 shows a 95.9% result, outperforming prior skeleton-based methods such as SkeletonMAE [23], PYSKL [6], and PoseConv3D [7]. Since FineGYM emphasizes fine-grained athletic motions, this result is consistent with the motivation of using prototypes to expose subtle local differences.

4.3. Component ablations

Table 3 studies the three main components on NTU-120 X-Sub. The baseline achieves 87.8%. Adding MTE improves performance to 88.1%, indicating that enhanced topology learning alone is beneficial. Adding CSC on top of MTE further increases accuracy to 88.3%, showing that class-specific supervision on the reconstructed topology space is helpful even without the full prototype pathway. Adding PRN to the MTE backbone yields 88.5%, and combining all three components reaches 89.0%, a total gain of 1.2 points over the baseline.

Table 4 provides two more targeted comparisons. On the left, ProtoGCN outperforms alternative graph-contrastive strategies adapted from SkeletonGCL [8] and FR-Head [26]. On the right, removing either adaptive topology branch reduces accuracy, and removing both causes the largest drop. This confirms that the topology enhancement and the prototype-regularized training objective are complementary rather than redundant.

5. Analysis

5.1. Where do the gains come from?

Table 5 groups NTU-120 classes by baseline difficulty. The gains are concentrated on the hardest classes: ProtoGCN improves the lowest-accuracy top-10 group by 3.0 points and the next 20 classes by 2.5 points, while the easiest classes improve only modestly. This pattern is important because it matches the intended effect of prototype reconstruction: the method helps most when coarse topology

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Table 1. Performance comparisons against state-of-the-art methods on the NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton datasets in terms of classification accuracy (%).

Methods	Publication	NTU RGB+D		NTU RGB+D 120		Kinetics-Skeleton	
		X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
ST-GCN [24]	AAAI 2018	81.5	88.3	70.7	73.2	30.7	52.8
AS-GCN [12]	CVPR 2019	86.8	94.2	78.3	79.8	34.8	56.5
2s-AGCN [18]	CVPR 2019	88.5	95.1	82.5	84.2	36.1	58.7
MS-G3D [15]	CVPR 2020	91.5	96.2	86.9	88.4	38.0	60.9
DC-GCN+ADG [4]	ECCV 2020	90.8	96.6	86.5	88.1	-	-
MST-GCN [3]	AAAI 2021	91.5	96.6	87.5	88.8	38.1	60.8
CTR-GCN [2]	ICCV 2021	92.4	96.8	88.9	90.6	-	-
STF [10]	AAAI 2022	92.5	96.9	88.9	89.9	39.9	-
InfoGCN [5]	CVPR 2022	93.0	97.1	89.8	91.2	-	-
PYSKL [6]	ACM MM 2022	92.6	97.4	88.6	90.8	49.1	-
SkeletonGCL [8]	ICLR 2023	92.8	97.1	89.8	91.2	-	-
FR-Head [26]	CVPR 2023	92.8	96.8	89.5	90.9	-	-
GAP [20]	ICCV 2023	92.9	97.0	89.9	91.1	-	-
HD-GCN [11]	ICCV 2023	93.4	97.2	90.1	91.6	40.9	63.5
JT-GraphFormer [25]	AAAI 2024	93.4	97.5	89.9	91.7	-	-
DS-GCN [21]	AAAI 2024	93.1	97.5	89.2	91.1	50.6	-
BlockGCN [27]	CVPR 2024	93.1	97.0	90.3	91.5	-	-
ProtoGCN (2-ensemble)		93.0	97.2	89.7	91.2	49.9	74.0
ProtoGCN (4-ensemble)		93.5	97.5	90.4	91.9	51.3	75.1
ProtoGCN (6-ensemble)		**93.8**	**97.8**	**90.9**	**92.2**	**51.9**	**75.6**

Table 2. Performance comparisons against state-of-the-art methods on FineGYM in terms of classification accuracy (%).

Methods	Modality	FineGYM
I3D [1]	RGB	64.4
TSN [19]	RGB+Flow	79.8
TSM [13]	RGB+Flow	81.2
LT-S3D [22]	RGB	88.9
SkeletonMAE [23]	Skeleton	91.8
PYSKL [6]	Skeleton	94.1
PoseConv3D [7]	Skeleton+Limb	94.3
ProtoGCN (Ours)	Skeleton	**95.9**

Table 3. Ablation study on the contribution of each component in ProtoGCN under the NTU-120 X-Sub setting.

Baseline	MTE	PRN	CSCL	Acc (%)
✓	–	–	–	87.8
✓	✓	–	–	88.1 (↑ 0.3)
✓	✓	–	✓	88.3 (↑ 0.5)
✓	✓	✓	–	88.5 (↑ 0.7)
✓	✓	✓	✓	**89.0 (↑ 1.2)**

Table 4. Comparison of classification accuracies (%) based on (*left*) existing graph contrastive methods and (*right*) different enhanced topologies for the network.

Methods	Acc (%)	Methods	Acc (%)
Baseline	87.8	ProtoGCN	**89.0**
w/ SkeletonGCL [8]	88.0	w/o $\mathbf{A}_{\mathbf{intra}}$	88.8
w/ FR-Head [26]	88.2	w/o $\mathbf{A}_{\mathbf{inter}}$	88.5
Ours	**89.0**	w/o $\mathbf{A}_{\mathbf{intra}}, \mathbf{A}_{\mathbf{inter}}$	88.4

Table 5. Average performance (%) on different difficulty level classes sorted by accuracy under the NTU-120 X-Sub setting.

Difficulty Level	Baseline	ProtoGCN	Δ
class 1-10	64.1	67.1	**+3.0**
class 11-30	78.6	81.1	**+2.5**
class 31-60	88.6	89.7	**+1.1**
class 61-120	96.0	96.4	**+0.4**

cues are insufficient and discrimination depends on small but meaningful relation changes.

Figure 3 compares topology visualizations between PYSKL [6] and ProtoGCN on four actions. ProtoGCN consistently produces more localized and interpretable empha-

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sis around the body parts that define the action, such as the head for *Nod Head*, the foot-leg area for *Wear a Shoe* and *Kick*, and the hand-arm area for *Drop*. In contrast, the baseline visualizations appear more diffuse. While qualitative, these results align with the quantitative improvements on difficult classes.

5.2. Hyperparameter sensitivity

Figure 4 studies the contrastive weight λ and the prototype memory capacity n_{pro} on NTU-120 X-Sub. Performance is stable for moderate λ values and peaks at $\lambda = 0.3$, while overly large contrastive weight degrades accuracy. The prototype memory also shows a clear trade-off: too few prototypes underfit, while excessively large memory does not help further. The best ablation result in the provided plot occurs at $n_{\text{pro}} = 100$.

5.3. Class-wise behavior

Figure 5 shows the action classes with accuracy differences above 1% between ProtoGCN and PYSKL on NTU-120. The strongest positive gains occur on actions whose semantics depend on subtle local motion or human-object interaction, such as *staple book*, *counting money*, *writing*, and *typing on a keyboard*. This is exactly the regime where a prototype-based topology representation is expected to help. The figure also shows a small set of negative transfers, for example on classes such as *touch head*, *make ok sign*, and *cutting nails*. These failures indicate that the method is not universally beneficial, but the negative changes are generally smaller in magnitude than the best positive gains. The remaining failure cases are also informative: they mostly involve actions with very short temporal extent or ambiguous hand poses, where the observed joints provide limited evidence even after topology enhancement. This suggests that prototype reconstruction is most effective when small but repeatable motion relations exist, and less effective when the ambiguity comes from missing pose detail rather than relational modeling.

6. Conclusion

We presented ProtoGCN, a prototype-driven framework for skeleton-based action recognition that reconstructs learned motion topologies from a prototype memory and regularizes them with a class-specific contrastive objective. The method is simple to integrate into a modern GCN backbone: MTE improves topology quality, PRN exposes reusable relation patterns, and CSC makes the reconstructed topology space more discriminative. Across NTU RGB+D, NTU RGB+D 120, Kinetics-Skeleton, and FineGYM, ProtoGCN achieves the best results among the provided comparisons. The analyses further show that the gains are concentrated on difficult and fine-grained classes, which supports the main claim that prototype-based reconstruction helps reveal subtle but decisive motion details.

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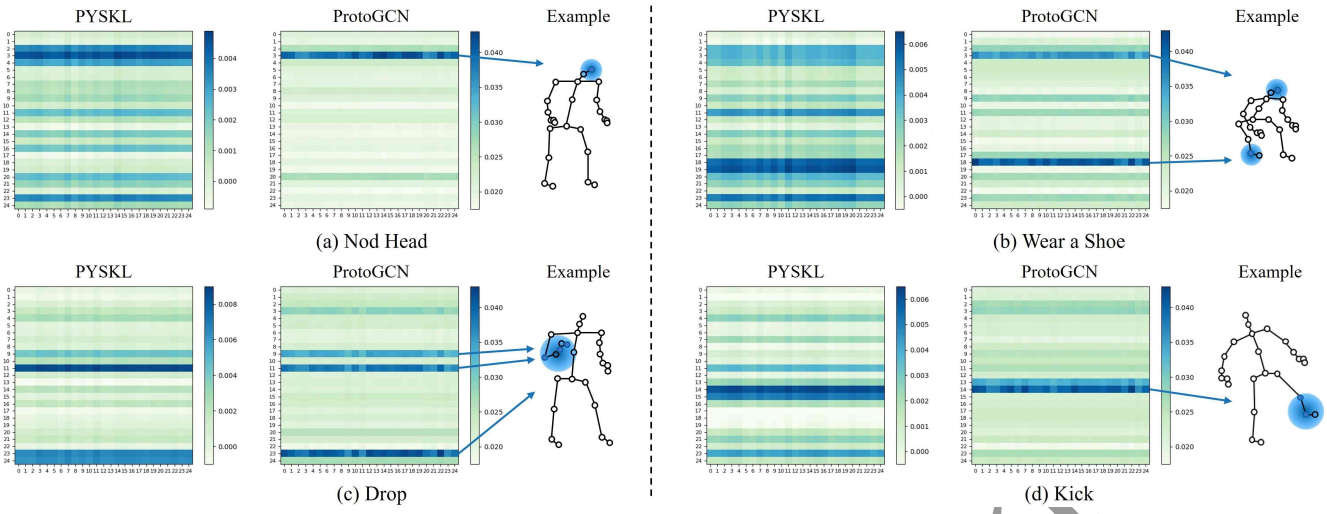

Figure 3. Topology visualization on four actions. Compared with PYSKL [6], ProtoGCN places stronger emphasis on action-defining body parts.

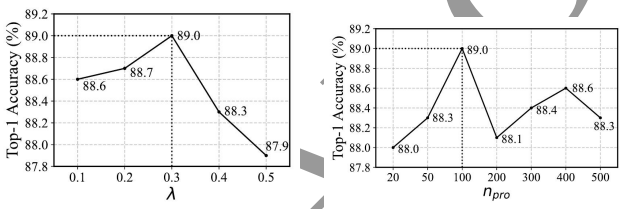

Figure 4. Sensitivity to the CSC loss weight λ and prototype count n_{pro} on NTU-120 X-Sub.

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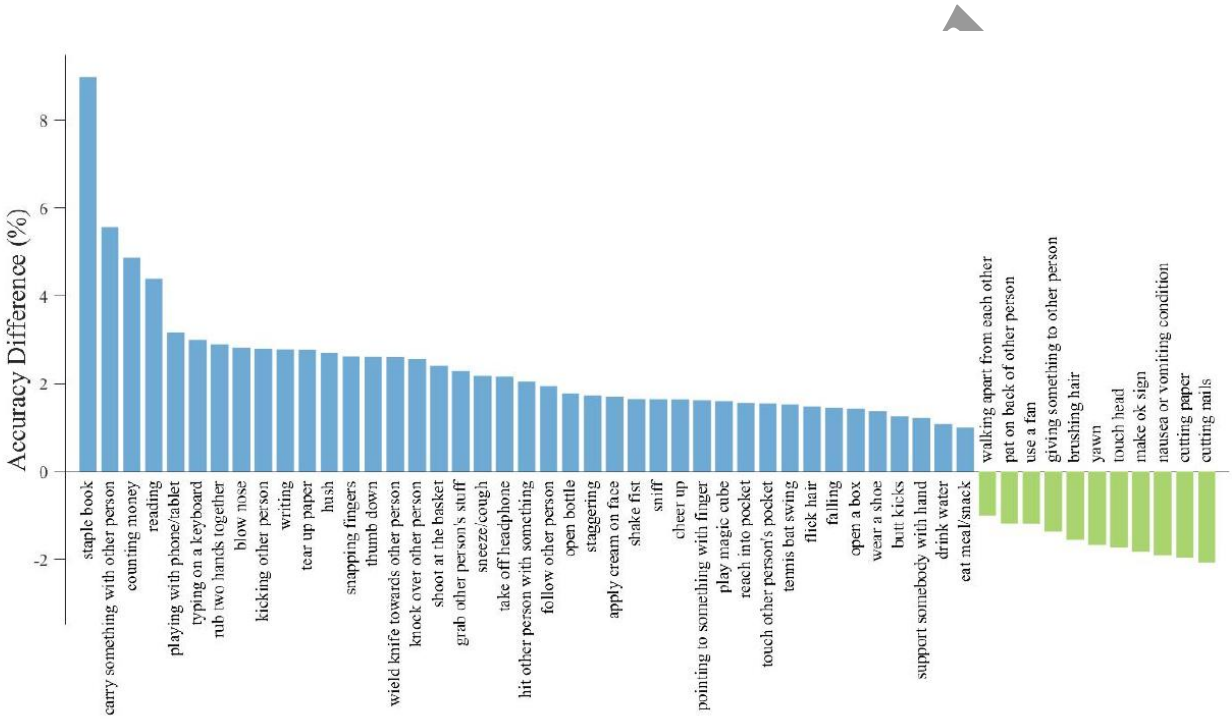

Figure 5. Class-wise accuracy difference between ProtoGCN and PYSKL [6] on NTU-120. The largest improvements appear on several fine-grained classes, while a smaller set of classes degrades slightly.

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